

## Lecture 2 : Sequence

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**Theorem 1** Let  $(u_n)$  and  $(v_n)$  be two sequences and  $\lim_{n \rightarrow \infty} u_n = u$  and  $\lim_{n \rightarrow \infty} v_n = v$ . Then

1.  $\lim(u_n + v_n) = u + v$ ;
2.  $\lim(u_n - v_n) = u - v$ ;
3.  $\lim(u_n v_n) = uv$ ;
4.  $\lim\left(\frac{u_n}{v_n}\right) = \frac{u}{v}$  provided  $(v_n)$  is a sequence of non zero real numbers and  $v \neq 0$ .

**Proof:**

1. To show that  $\lim(u_n + v_n) = u + v$ , we need to establish that for a preassigned positive number  $\epsilon$ , there exists a natural number  $k$  such that

$$|(u_n + v_n) - (u + v)| < \epsilon \text{ for all } n \geq k$$

Using triangle inequality, we have

$$|(u_n + v_n) - (u + v)| = |(u_n - u) + (v_n - v)| < |u_n - u| + |v_n - v|.$$

Let  $\epsilon > 0$ . Since,  $\lim u_n = u$ , there exists a natural number  $k_1$  such that  $|u_n - u| < \frac{\epsilon}{2}$  for all  $n \geq k_1$ .

Since,  $\lim v_n = v$ , there exists a natural number  $k_2$  such that  $|v_n - v| < \frac{\epsilon}{2}$  for all  $n \geq k_2$ .

Let  $k = \max\{k_1, k_2\}$ . Then  $|u_n - u| < \frac{\epsilon}{2}$  and  $|v_n - v| < \frac{\epsilon}{2}$  for all  $n \geq k$ . It follows that  $|(u_n + v_n) - (u + v)| < \epsilon$  for all  $n \geq k$ .

Since  $\epsilon$  is arbitrary and hence  $\lim(u_n + v_n) = u + v$ .

2. To show that  $\lim(u_n - v_n) = u - v$ , we need to establish that for a preassigned positive number  $\epsilon$ , there exists a natural number  $k$  such that

$$|(u_n - v_n) - (u - v)| < \epsilon \text{ for all } n \geq k$$

Now, we have

$$|(u_n - v_n) - (u - v)| = |(u_n - u) - (v_n - v)| < |u_n - u| + |v_n - v|.$$

Let  $\epsilon > 0$ . Since,  $\lim u_n = u$ , there exists a natural number  $k_1$  such that  $|u_n - u| < \frac{\epsilon}{2}$  for all  $n \geq k_1$ .

Since,  $\lim v_n = v$ , there exists a natural number  $k_2$  such that  $|v_n - v| < \frac{\epsilon}{2}$  for all  $n \geq k_2$ .

Let  $k = \max\{k_1, k_2\}$ . Then  $|u_n - u| < \frac{\epsilon}{2}$  and  $|v_n - v| < \frac{\epsilon}{2}$  for all  $n \geq k$ .  $|(u_n - v_n) - (u - v)| < \epsilon$  for all  $n \geq k$ .

Since,  $\epsilon$  is arbitrary and hence  $\lim(u_n - v_n) = u - v$ .

3. To show that  $\lim(u_n v_n) = uv$ , we need to establish that for a preassigned positive number  $\epsilon$  there exists a natural number  $k$  such that

$$|u_n v_n - uv| < \epsilon \text{ for all } n \geq k.$$

we know that,

$$|u_n v_n - uv| = |u_n(v_n - v) + v(u_n - u)| \leq |u_n| |v_n - v| + |v| |u_n - u|.$$

Since,  $(u_n)$  is a convergent sequence and so it is bounded. Therefore, there exists a positive number  $B_1$  such that  $|u_n| \leq B_1$  for all  $n \in \mathbb{N}$ .

Let  $B = \max\{B_1, |v|\}$ .

Then,  $|u_n v_n - uv| < B|v_n - v| + B|u_n - u|$ .

Let  $\epsilon > 0$ . Since,  $\lim u_n = u$ , there exists a natural number  $k_1$  such that  $|u_n - u| < \frac{\epsilon}{2B}$  for all  $n \geq k_1$ .

Since,  $\lim v_n = v$ , there exists a natural number  $k_2$  such that  $|v_n - v| < \frac{\epsilon}{2B}$  for all  $n \geq k_2$ .

Let  $k = \max\{k_1, k_2\}$ . Then,  $|u_n - u| < \frac{\epsilon}{2B}$  and  $|v_n - v| < \frac{\epsilon}{2B}$  for all  $n \geq k$ .

It follows that  $|u_n v_n - uv| < B \cdot \frac{\epsilon}{2B} + B \cdot \frac{\epsilon}{2B}$  for all  $n \geq k$  or  $|u_n v_n - uv| < \epsilon$  for all  $n \geq k$ . Since,  $\epsilon$  is arbitrary and therefore  $\lim(u_n v_n) = uv$ .

4. First we prove that if  $\lim v_n = v$  where  $(v_n)$  is a sequence of non-zero real numbers and  $v \neq 0$ ,  $\lim(1/v_n) = 1/v$ .

Let  $\alpha = \frac{1}{2}|v|$ . Then,  $\alpha > 0$ . Since,  $\lim v_n = v$ , there exists a natural number  $k_1$  such that  $|v_n - v| < \alpha$  for all  $n \geq k_1$ .

We have  $||v_n| - |v|| \leq |v_n - v| < \alpha$  for all  $n \geq k_1$  or  $|v| - \alpha < |v_n| < |v| + \alpha$  for all  $n \geq k_1$ .

Therefore,  $|v_n| > \frac{1}{2}|v|$  for all  $n \geq k_1$ .

$$\left| \frac{1}{v_n} - \frac{1}{v} \right| = \frac{|v_n - v|}{|v| |v_n|} < \frac{2}{|v|^2} |v_n - v| \text{ for all } n \geq k_1.$$

Let  $\epsilon > 0$ . Since,  $\lim v_n = v$ , there exists a natural number  $k_2$  such that  $|v_n - v| < \frac{|v|^2}{2} \epsilon$  for all  $n \geq k_2$ .

Let  $k = \max\{k_1, k_2\}$ . Then,  $\left| \frac{1}{v_n} - \frac{1}{v} \right| < \epsilon$  for all  $n \geq k$ .

Since,  $\epsilon$  is arbitrary and so  $\lim \frac{1}{v_n} = \frac{1}{v}$ . The proof of the theorem is now completed by considering the convergence of the product of two sequences  $(u_n)$  and  $(\frac{1}{v_n})$ .

Therefore,  $\lim \left( \frac{u_n}{v_n} \right) = \lim \left( u_n \cdot \frac{1}{v_n} \right) = u \cdot \frac{1}{v} = \frac{u}{v}$ .

## 1. Divergent sequence

A real sequence  $(f(n))$  is said to diverge to  $\infty$  if corresponding to a preassigned positive number  $G$ , however large, there exists a natural number  $k$  such that

$$f(n) > G \text{ for all } n \geq k.$$

In this case, we write  $\lim f(n) = \infty$  and also say that the sequence  $(f(n))$  tends to  $\infty$ . A real sequence  $(f(n))$  is said to diverge to  $-\infty$  if corresponding to a preassigned positive number  $G$ , however large, there exists a natural number  $k$  such that

$$f(n) < -G \text{ for all } n \geq k.$$

In this case we write  $\lim f(n) = -\infty$  and also say that the sequence  $(f(n))$  tends to  $-\infty$ .

### Examples

1. The sequence  $2^n$ ,  $n \in \mathbb{N}$  diverges to  $\infty$ .
2. The sequence  $-n^2$ ,  $n \in \mathbb{N}$  diverges to  $-\infty$ .

### 2. Oscillating sequence

A sequence  $(f(n))$  is said to be **oscillating sequence** if it is neither convergent nor divergent. Oscillating sequence are of two types :

1. A sequence is said to be **finitely oscillating** if it is bounded as well as oscillating.
2. A sequence is said to be **infinitely oscillating** if it is unbounded as well as oscillating.

### Examples

1. The sequence  $((-1)^n)$ ,  $n \in \mathbb{N}$  bounded and oscillating. Hence it is an oscillating sequence of finite oscillation.
2. The sequence  $((-1)^n n)$ ,  $n \in \mathbb{N}$ . unbounded and oscillating. Hence it is an oscillating sequence of infinite oscillation.

### 3. Monotone sequence

A real sequence  $(f(n))$  is said to be a **monotone increasing sequence** if  $f(n+1) \geq f(n)$  for all  $n \in \mathbb{N}$ .

A real sequence  $(f(n))$  is said to be a **monotone decreasing sequence** if  $f(n+1) \leq f(n)$  for all  $n \in \mathbb{N}$ .

A real sequence  $(f(n))$  is said to be a **monotone sequence** if it is either a monotone increasing sequence or a monotone decreasing sequence.

**Note :** If  $f(n+1) > f(n)$  for all  $n \in \mathbb{N}$  the sequence  $(f(n))$  is said to be a **strictly monotone increasing sequence**.

If  $f(n+1) < f(n)$  for all  $n \in \mathbb{N}$  the sequence  $(f(n))$  is said to be a **strictly monotone decreasing sequence**.

## Examples

1. Let  $f(n) = 2^n$ ,  $n \geq 1$ .  
Then,  $f(n+1) > f(n)$  for all  $n \in \mathbb{N}$ .  
Therefore, the sequence  $(f(n))$  is a monotone increasing sequence. It is also strictly monotone.
2. Let  $f(n) = \frac{1}{n}$ ,  $n \geq 1$ .  
Then,  $f(n+1) < f(n)$  for all  $n \in \mathbb{N}$ .  
Therefore, the sequence  $(f(n))$  is a monotone decreasing sequence. It is also strictly monotone.
3. The sequence  $((-2)^n)$  is neither a monotone increasing sequence nor a monotone decreasing sequence. Therefore, it is not a monotone sequence.

**Theorem 2** *Every monotonic increasing and bounded above sequence is convergent and it converges to the least upper bound.*

**Proof:** Let  $(f(n))$  be a monotone increasing sequence bounded above and let  $M$  be its least upper bound. Then

1.  $f(n) \leq M$  for all  $n \in \mathbb{N}$  and
2. for preassigned positive number  $\epsilon$ , there exists a natural number  $k$  such that  $f(k) > M - \epsilon$ .

Since,  $(f(n))$  is a monotone increasing sequence,

$$M - \epsilon < f(k) \leq f(k+1) \leq f(k+2) \leq \dots \leq M.$$

This implies,  $M - \epsilon < f(n) < M + \epsilon$  for all  $n \geq k$ , i.e.  $|f(n) - M| < \epsilon$  for all  $n \geq k$ .  
This shows that the sequence  $(f(n))$  is convergent and  $\lim f(n) = M$ .

**Theorem 3** *Every monotonic decreasing and bounded below sequence is convergent and it converges to the greatest lower bound.*

**Proof:** Let  $(f(n))$  be a monotone decreasing sequence bounded below and let  $m$  be its greatest lower bound. Then

1.  $f(n) \geq m$  for all  $n \in \mathbb{N}$  and
2. for preassigned positive number  $\epsilon$ , there exists a natural number  $k$  such that  $f(k) < m + \epsilon$ .

Since,  $(f(n))$  is a monotone decreasing sequence,

$$m - \epsilon \leq \dots \leq f(k+2) \leq f(k+1) \leq f(k) < m + \epsilon.$$

This implies,  $m - \epsilon < f(n) < m + \epsilon$  for all  $n \geq k$ , i.e.  $|f(n) - m| < \epsilon$  for all  $n \geq k$ .  
This shows that the sequence  $(f(n))$  is convergent and  $\lim f(n) = m$ .

#### 4. Cauchy sequence

A sequence  $(u_n)$  is said to be a Cauchy sequence if for a preassigned positive number  $\epsilon$ , there exists a natural number  $k$  such that

$$|u_m - u_n| < \epsilon \text{ for all } m, n \geq k.$$

Replacing  $m$  by  $n + p$  where  $p = 1, 2, 3, \dots$  the above condition can be equivalently stated as

$$|u_{n+p} - u_n| < \epsilon \text{ for all } n \geq k \text{ and } p = 1, 2, 3, \dots$$

**Note :** A sequence is convergent if and only if it is Cauchy sequence.  
(For proof, see the reference book)

**Reference book :** *Introduction to real analysis by S.K. Mapa.*